Ayush Kumar

Robotics Software-Electronics Engineer — System Integrations

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Education

Indian Institute of Information Technology, Design & Manufacturing, Kancheepuram **2022-2026**
B.Tech Electronics & Communication Engineering *Chennai, India*

Achievements

- **International Rover Challenge 2025:** Led a 50+ member team to 16th place internationally.
- **ISRO Robotics Challenge:** Secured 6th place nationwide among 267 teams, recognized by the President of India on National Space Day.
- **NSO International Computer Science Olympiad:** Secured international rank 32 and national rank 16.

Experience

Virya Autonomous Technologies
- Robotic Software Intern *07/2025 - 12/2025*

- **High-Fidelity State Estimation:** Benchmarked and tuned Cartographer and RTAB-Map for an Autonomous Mobile Robot (AMR) against RTK ground truth, achieving highly accurate localization with an Absolute Trajectory Error (ATE) of < 20 cm and Relative Pose Error (RPE) of < 10 cm.
- **Odometry & Compute Optimization:** Re-architected the Direct LiDAR-Inertial Odometry (DLIO) pipeline by implementing dynamic queue management, successfully slashing average RAM consumption from ∼4.5 GB to 470 MB without sacrificing real-time performance.
- **Advanced SLAM Architecture:** Optimized RTAB-Map computational efficiency and reliability for state estimation by integrating Fast-GICP and evaluating various graph optimizers (g2o, GTSAM) and CV feature extractors (ORB, FAST, SIFT).
- **Perception & Sensor Fusion:** Integrated a multi-modal sensor suite (3D LiDAR, IMU, depth cameras) over stable DDS profiles. Engineered a custom point cloud pipeline for data compression and mathematically projected 3D LiDAR data into equirectangular and simulated multi-camera pinhole views.
- **Navigation Architecture & Routing:** Developed a custom JavaScript-based ROS environment to streamline navigation stack deployment, featuring interactive PCD and grid map editing. Rewrote the core routing algorithm for predefined paths and implemented custom behavior layers for robust autonomous task execution.

Hyper Horizon
- Intern - Autonomous Systems *05/2025 - 06/2025*

- **Advanced State Estimation:** Engineered a robust multi-sensor state estimation pipeline by developing and tuning Extended and Unscented Kalman Filters (EKF/UKF) to fuse multi-modal data. This ensured high navigational accuracy for the AUV, especially during transitions into GPS-denied underwater environments.
- **Sensor Integration & Calibration:** Led the physical integration and rigorous calibration of a complex subsea sensor suite—including DVL, AHRS, RTK GPS, and IMU—for a defense-grade Autonomous Underwater Vehicle (AUV).
- **Autonomous System Development:** Contributed to the development of a complete autonomous navigation and control system, bridging the gap from low-level sensor integration up to high-level mission execution.
- **Lifecycle & Environment Architecture:** Streamlined the end-to-end robotics software lifecycle by architecting remote development environments. Deployed C++ control algorithms, conducted comprehensive simulation-based tuning, and integrated a React-based operator GUI for system monitoring.

Mars Rover Students Club, IIITDM
- Team Leader *04/2024 - 02/2025*

- **Cross-Functional Leadership:** Led a 30+ member team in ISRO's IROC-U24, successfully securing 6th place nationwide among 267 teams. This achievement was recognized by the President of India on National Space Day.
- **End-to-End System Integration:** Oversaw the comprehensive system integration across all mechanical, electrical, and software subsystems. Acted as the technical bridge, architecting solutions from low-level embedded systems up to high-level navigation controls.
- **Robust Control & Planning Stack:** Integrated and fine-tuned a diverse suite of global and local path planners (A*, Hybrid A*) alongside advanced controllers (MPPI, Pure Pursuit, DWA).
- **Autonomous Terrain Navigation:** Engineered the navigation pipeline to enable the rover to reliably traverse rough, unstructured terrain while executing dynamic obstacle avoidance.

Projects

Hospitality Co-Bot

- **Custom Control Architecture:** Developed a custom UDP-based hardware interface plugin within the ROS 2 Humble `ros2_control` framework, enabling precise closed-loop PID control of a differential-drive base (`diff_drive_controller`) on NVIDIA Jetson edge hardware.
- **2D Mapping & Localization:** Implemented a robust 2D SLAM pipeline utilizing Cartographer and SLAM Toolbox for high-fidelity environment mapping and reliable pose estimation against pre-existing floor plans.
- **Dynamic Navigation Stack:** Integrated the Nav2 stack with custom Behavior Trees and 2D costmap plugins, tailoring the system for safe, dynamic obstacle avoidance in unstructured, human-centric environments.
- **Simulation & Hardware Validation:** Designed and validated the system's kinematics and URDF within a Gazebo simulation, bridging the gap between simulated motion planning and the physical UDP-based hardware deployment.

- Reinforcement Learning-Based Quadruped (Spider Bot)

Ongoing Independent Project

- **RL-Driven Locomotion:** Architecting a reinforcement learning framework to govern the complex kinematics and autonomous locomotion of a custom four-legged (quadrupedal) robot.
- **Kinematic Modeling & Control:** Currently formulating the state-action space and reward functions to train isolated single-leg actuation, establishing the foundational policy for full-body gait generation.
- **Simulation & Training Environment:** Developing the initial simulation parameters to iteratively train the agent, with the overarching goal of achieving stable, point-to-point autonomous navigation.

- 6-DOF Manipulator

- **End-to-End System Design:** Architected and developed a custom mobile manipulator UGV from the ground up, specifically targeting autonomous pick-and-place operations within unstructured environments.
- **Kinematics & Control Stack:** Engineered the complete autonomous control pipeline for the 6-DOF robotic arm utilizing ROS 1 and MoveIt.
- **Collision-Free Trajectory Planning:** Implemented and rigorously benchmarked Open Motion Planning Library (OMPL) algorithms, including RRT and RRTConnect, to generate optimal, collision-free motion trajectories.
- **Physics-Based Simulation:** Constructed a high-fidelity Gazebo simulation featuring a meticulously tuned URDF. This enabled precise validation of the manipulator's kinematics and dynamic behavior prior to physical hardware deployment.

Technical Skills

Languages: C++, Python, Embedded C, Bash/Shell, MATLAB, JavaScript

Robotics Core: ROS 1 & 2, Nav2, MoveIt, TF2, URDF/Xacro, Gazebo, Rviz, `ros2_control` (Humble), DDS Profiles

Algorithms & Control: SLAM (Cartographer, RTAB-Map, DLIO), SLAM Toolbox, FAST-LIO, Graph Optimization (g2o, GTSAM), Fast-GICP, Path Planning (A*, RRT, MPPI, DWA), Hybrid A*, RRTConnect, Pure Pursuit, State Estimation (EKF, UKF), Closed-Loop PID Control, Reinforcement Learning

Embedded, OS & HW: Linux (Ubuntu, Debian), Yocto Project, ARM Architecture, Real-Time Systems, Docker, NVIDIA Jetson (Xavier, Nano, Orin Nano), STM32 Microcontrollers

Libraries & Perception: PCL (Point Cloud Library), OpenCV, Eigen, OMPL, Ceres Solver, Computer Vision (ORB, FAST, SIFT), Sensor Integration (LiDAR, IMU, DVL, AHRS, RTK GPS, Intel RealSense)